UNIT 2: DATABASE NORMALIZATION & FUNCTIONAL DEPENDENCY

Course: CS301 Database Systems | Department of Computer Science

1. Introduction to Schema Refinement

In relational database design, the primary goal of schema refinement is to avoid redundancy.
Uncontrolled redundancy leads to three major update anomalies:
- Insertion Anomaly: Inability to store certain information without adding unrelated dummy data.
- Deletion Anomaly: Loss of crucial information when unrelated records are deleted.
- Modification/Update Anomaly: Inconsistencies arising when updating duplicated data records.

2. Functional Dependencies (FDs)
A functional dependency is a formal constraint between two sets of attributes in a relation R.
Formally denoted as X -> Y, where X is called the Determinant and Y is called the Dependent.
This condition asserts that for any two tuples t1 and t2 in R, if t1[X] = t2[X], then t1[Y] must equal t2[Y].

2. Normal Forms Hierarchy (1NF, 2NF, 3NF)

First Normal Form (1NF):
A relation R is in 1NF if and only if all attribute values are atomic and indivisible.
Multi-valued attributes and composite attributes are strictly prohibited.

Second Normal Form (2NF):
A relation R is in 2NF if it is already in 1NF and every non-prime attribute is fully functionally dependent on the primary key (no partial functional dependency exists).
A partial dependency occurs when a non-key attribute depends on a proper subset of candidate key.

Third Normal Form (3NF):
A relation R is in 3NF if it is in 2NF and there is no transitive dependency of non-prime attributes on candidate keys. For every non-trivial FD X -> Y, either:

1. X is a superkey of R, OR
2. Y is a prime attribute (part of some candidate key).

3. Boyce-Codd Normal Form (BCNF) & Decomposition Properties

Boyce-Codd Normal Form (BCNF):
BCNF is a stricter variant of 3NF. A relation R is in BCNF if for every non-trivial functional dependency X -> Y, X MUST be a superkey.

Decomposition Theorems:
When decomposing relation R into R1 and R2, two crucial properties must be tested:

1. Lossless Join Decomposition:
   Natural join of R1 and R2 must reconstruct original relation R without spurious tuples.
   Condition: (R1 · R2 -> R1) OR (R1 · R2 -> R2)
2. Dependency Preservation:
   All functional dependencies of R must be logically implied by the union of projected FDs.